Data Transformation in Power Query - Assignment Instruction

Task 1: Merging multiple tables together

Data To be Used:

- Merging Tables – Start.xlsx

There are three sheets in the above excel files as follows:

- Employee – this is a table with all the Sales employees information
- BusinessEntity – each employee is assigned to a specific business entity in their region. This is a reference table for all those business entities in every region.
- Sales – this table holds all the sales made by each of the sales employees

Objective:

- Merge the Employee and Business entity and Sales tables and group by (business entity, region and Employee ID) for all rows.

- The desired result table should be structured as

EmployeeID	BusinessEntity	FirstName	LastName	RegionBE	Sum of OrderQty	Sum of LineTotal	TerritoryID
501	3	Mindy	Martin	BE_248STW - 45	0	0	7
502	2	Frank	Martinez	BE_314STW - 47	0	0	4
503	7	Steve	Masters	BE_535STW - 41	0	0	6
504	3	Joseph	Matthews	BE_248STW - 45	0	0	7
505	10	Jennifer	Maxham	BE_536STW - 38	0	0	10
506	1	Chris	Maxwell	BE_521STW - 49	0	0	1
507	7	Sandra	Maynard	BE_535STW - 37	0	0	4
508	8	Walter	Mays	BE_371STW - 31	0	0	4
509	7	Katie	McAskill-White	BE_535STW - 32	0	0	2
510	10	Lola	McCarthy	BE_536STW - 48	0	0	1
511	4	Jane	McCarty	BE_518STW - 42	0	0	4
512	8	John	McClane	BE_371STW - 46	0	0	5
513	5	Nikki	McCormick	BE_273STW - 43	0	0	5
514	2	James	McCoy	BE_314STW - 37	0	0	6
515	4	Christinia	McDonald	BE_518STW - 42	0	0	8
516	1	Alejandro	McGuel	BE_521STW - 47	0	0	2
517	6	Robin	McGuigan	BE_367STW - 49	0	0	6
518	5	Yvonne	McKay	BE_273STW - 46	0	0	3
519	10	Nkenge	McLin	BE_536STW - 48	0	0	1
520	2	Nancy	McPhearson	BE_314STW - 40	0	0	10

Task2: Grouping Multiple Rows in a Table

Data To be Used:

- Group by – Start.xlsx”

There is only a single sheet in the above excel file as:

- Sales – daily information of sales by product and Channel Name

Objective:

Add following columns to the source table using group by technique in Power Query

- Total Sales: add a column with total amount by Date and Channel.
- Total Products: add a column with unique products sold on each Date and Channel.
- Products Sold: Create a new column with a list of all the products sold on that day (separated by commas).
- Top Product: Create another column containing the product with the highest sales on each date.
- Top Product Participation: Create a final column with the percentage share of total sales for the top selling product

Existing data structure is as follows

Date	ProductName	ChannelName	Amount
14-12-2016	Rob Collie's shirt	Reseller	86.35
09-10-2016	Custom Sleeve	Reseller	11.41
11-11-2016	Sleeveless	Online	30.76
25-09-2016	Sleeveless	Reseller	166.08
14-10-2016	Cap Sleeve	Online	178.44
04-12-2016	Ken's Special Slee	Online	396.94
07-10-2016	Cap Sleeve	Online	126.2

The desired results are as follows

Date	Channel Name	Products Sold	Top Product Participation	Total Sales	Total Products	Top Product
27-09-2016	Store	Rob Collie's shirt	100.00%	34.86	1	Rob Collie's shirt
24-10-2016	Reseller	Cap Sleeve	100.00%	73.25	1	Cap Sleeve
28-09-2016	Reseller	Long Sleeve	100.00%	460.1	1	Long Sleeve
22-09-2016	Online	3/4 Sleeve	100.00%	156.7	1	3/4 Sleeve
16-10-2016	Online	3/4 Sleeve	100.00%	160.9	1	3/4 Sleeve
04-10-2016	Store	Custom Sleeve	100.00%	44.64	1	Custom Sleeve
04-12-2016	Reseller	Sleeveless	100.00%	300.21	1	Sleeveless
13-12-2016	Reseller	Custom Sleeve	100.00%	51.46	1	Custom Sleeve
08-10-2016	Online	Short Sleeve	100.00%	70.84	1	Short Sleeve
05-12-2016	Reseller	Long Sleeve	100.00%	47.12	1	Long Sleeve
16-09-2016	Reseller	Rob Collie's shirt	100.00%	17.09	1	Rob Collie's shirt
07-10-2016	Reseller	Long Sleeve	100.00%	222.28	1	Long Sleeve
23-09-2016	Reseller	Custom Sleeve	100.00%	126.23	1	Custom Sleeve
16-10-2016	Reseller	Short Sleeve	100.00%	91.87	1	Short Sleeve

Task3: Combining Files from a Folder Data

To be Used:

- All files in the “Sales Reports” folder

Objective:

We have six Excel workbooks, one file for each month running from January to June. In each file, we only have one worksheet with the appropriate data for the corresponding month. We would like to:

- Combine all the files into one tall table
- Do some needed transformation before consolidating the files

Existing data structure is as follows

Month	Name	Australia	Canada	Central	France	Germany	Northeast	Northwest	Southeast	Southwest	United Kingdom
January	All-Purpose Bike Stand	$3,680.15	$58,477.61	$22,201.41	$51,053.63	$5,509.38	$25,223.52	$14,048.58	$50,065.79	$37,969.13	$9,236.07
	AWC Logo Cap	$2,154.54	$20,524.61	$53,371.81	$19,970.41	$20,627.03	$19,179.70	$22,498.86	$4,122.20	$53,854.45	$6,687.86
	Bike Wash - Dissolver	$1,157.14	$80,566.97	$3,884.30	$1,001.19	$31,448.81	$3,552.03	$31,177.21	$17,753.13	$50,414.38	$2,722.50
	Chain	$23,393.15	$3,746.69	$60,366.84	$61,502.97	$34,251.43	$813.16	$2,657.37	$16,895.67	$10,970.60	$1,644.17
	Classic Vest, M	$782.56	$1,487.69	$4,048.33	$10,384.59	$22,730.10	$15,236.66	$2,566.96	$2,617.94	$628.38	$1,313.51
	Front Derailleur	$46,353.15	$64,576.29	$3,004.76	$50,428.18	$31,031.19	$3,705.50	$4,870.95	$491.21	$3,025.79	$42,530.32
	Full-Finger Gloves, M	$18,929.68	$5,729.84	$8,445.64	$17,952.72	$36,642.71	$316.48	$20,871.13	$2,811.42	$10,746.33	$14,278.99
	Hitch Rack - 4-Bike	$24,929.13	$66,102.26	$52,112.95	$2,768.55	$20,631.18	$982.32	$12,865.55	$23,136.59	$7,234.92	$53,930.80
	HL Fork	$4,679.26	$24,684.26	$64,145.68	$14,616.43	$7,801.10	$3,524.28	$14,945.39	$59,482.66	$21,711.41	$1,841.39
	HL Headset	$3,758.19	$58,842.08	$12,214.49	$12,309.35	$25,577.32	$26,602.10	$30,589.99	$14,672.52	$35,854.92	$39,093.04
	HL Mountain Frame - Black, 44	$743.70	$3,129.67	$53,269.49	$15,492.02	$3,276.37	$11,997.59	$3,138.82	$44,837.22	$4,458.58	$73,133.96
	HL Mountain Frame - Silver, 42	$9,600.84	$37,099.83	$6,317.90	$8,918.88	$9,815.04	$23,613.40	$21,147.07	$11,220.14	$19,297.34	$2,894.56
	HL Mountain Frame - Silver, 48	$19,957.37	$170.48	$32,367.78	$17,340.92	$37,618.53	$21,356.91	$1,106.68	$33,263.77	$4,213.44	$14,804.82
	HL Mountain Front Wheel	$2,752.92	$2,777.33	$94.45	$4,940.28	$8,752.30	$15,325.30	$729.28	$26,290.47	$2,169.94	$21,175.46
	HL Mountain Handlebars	$32,956.90	$17,311.25	$21,978.25	$11,681.09	$65,574.29	$24,876.19	$11,820.27	$6,597.42	$33,915.11	$2,414.97
	HL Mountain Pedal	$17,613.41	$13,471.16	$38,674.14	$27,727.95	$16,550.44	$10,386.76	$6,097.75	$12,972.71	$14,780.65	$45,281.39
	HL Mountain Seat/Saddle	$24,943.39	$50,212.52	$1,589.56	$21,468.42	$1,753.25	$36,222.79	$4,321.54	$22,880.32	$40,521.41	$13,627.02
	HL Road Frame - Black, 48	$4,755.98	$7,577.35	$30,636.91	$9,273.79	$17,981.87	$23,989.75	$8,160.30	$15,259.48	$3,121.66	$3,937.16
	HL Road Frame - Red, 44	$5,981.09	$19,205.74	$1,386.59	$21,711.34	$389.84	$64,666.62	$2,120.76	$62,718.83	$3,421.48	$997.89
	HL Road Frame - Red, 48	$31,871.32	$23,152.58	$22,023.82	$38,002.10	$20,217.19	$14,931.31	$9,780.33	$10,298.94	$33,653.09	$38,398.87
	HL Road Frame - Red, 62	$1,968.79	$39,832.58	$24,220.62	$29,426.11	$16,801.15	$3,544.13	$14,644.87	$40,768.97	$44,705.45	$34,277.01
	HL Road Front Wheel	$821.45	$56,282.49	$51,739.26	$1,713.62	$515.94	$9,859.30	$870.54	$4,700.39	$347.39	$37,711.21
	HL Road Pedal	$66,744.86	$53,913.63	$6,040.41	$50,023.92	$43,361.23	$103.24	$7,406.78	$1,450.35	$38,620.77	$1,555.64

The desired results are as follows

	Month	Country	Name	Sales
1	January	Australia	All-Purpose Bike Stand	3680.15
2	January	Canada	All-Purpose Bike Stand	58477.61
3	January	Central	All-Purpose Bike Stand	22201.41
4	January	France	All-Purpose Bike Stand	51053.63
5	January	Germany	All-Purpose Bike Stand	5509.38
6	January	Northeast	All-Purpose Bike Stand	25223.52
7	January	Northwest	All-Purpose Bike Stand	14048.58
8	January	Southeast	All-Purpose Bike Stand	50065.79
9	January	Southwest	All-Purpose Bike Stand	37969.13
10	January	United Kingdom	All-Purpose Bike Stand	9236.07
11	January	Australia	AWC Logo Cap	2154.54
12	January	Canada	AWC Logo Cap	20524.61
13	January	Central	AWC Logo Cap	53371.81
14	January	France	AWC Logo Cap	19970.41
15	January	Germany	AWC Logo Cap	20627.03
16	January	Northeast	AWC Logo Cap	19179.7
17	January	Northwest	AWC Logo Cap	22498.86
18	January	Southeast	AWC Logo Cap	4122.2
19	January	Southwest	AWC Logo Cap	53854.45
20	January	United Kingdom	AWC Logo Cap	6687.86
21	January	Australia	Bike Wash - Dissolver	1157.14
22	January	Canada	Bike Wash - Dissolver	80566.97
23	January	Central	Bike Wash - Dissolver	3884.3
24	January	France	Bike Wash - Dissolver	1001.19